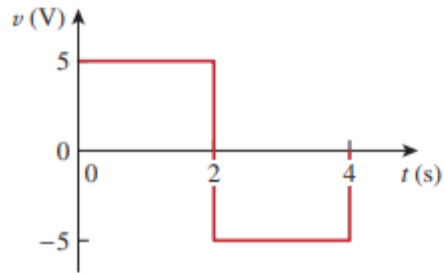
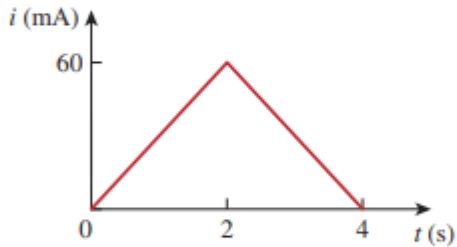


1.16 Figure 1.27 shows the current through and the voltage across an element.

- (a) Sketch the power delivered to the element for $t > 0$.
- (b) Find the total energy absorbed by the element for the period of $0 < t < 4$ s.



For $0 \leq t \leq 2$ s,

$$i(t) = \frac{60 - 0}{2 - 0} t = 30t = 30 \times 10^{-3} t \text{ (A)}$$

$$v(t) = 5V$$

$$p(t) = v(t)i(t) = 5 \times 0.030t = 0.15t \text{ Watt}$$

For $2 \leq t \leq 4$ s,

$$i(t) - i_2 = \frac{i_2 - i_1}{t_2 - t_1} (t - t_2)$$

$$i(t) - 0 = \frac{0 - 60}{4 - 2} (t - 4) = -30t + 120 \text{ (mA)}$$

$$i(t) = -30t + 120 \text{ (mA)} = -30 \times 10^{-3} t + 120 \times 10^{-3} \text{ (A)}$$

$$v(t) = -5V$$

$$p(t) = v(t)i(t) = -5 \times [-0.03t + 0.12] = [0.15t - 0.60] \text{ Watt}$$

How to sketch..

$$p(t) = 0.15t \text{ Watt, For } 0 \leq t \leq 2s,$$

$$@t = 0, p(t) = 0.15 \times 0 = 0$$

$$@t = 2, p(t) = 0.15 \times 2 = 0.3$$

$$p(t) = [0.15t - 6.0] \text{ Watt, For } 2 \leq t \leq 4s$$

$$@t = 2, p(t) = [0.15 \times 2 - 0.60] = -0.3$$

$$@t = 4, p(t) = [0.15 \times 4 - 0.60] = 0$$

